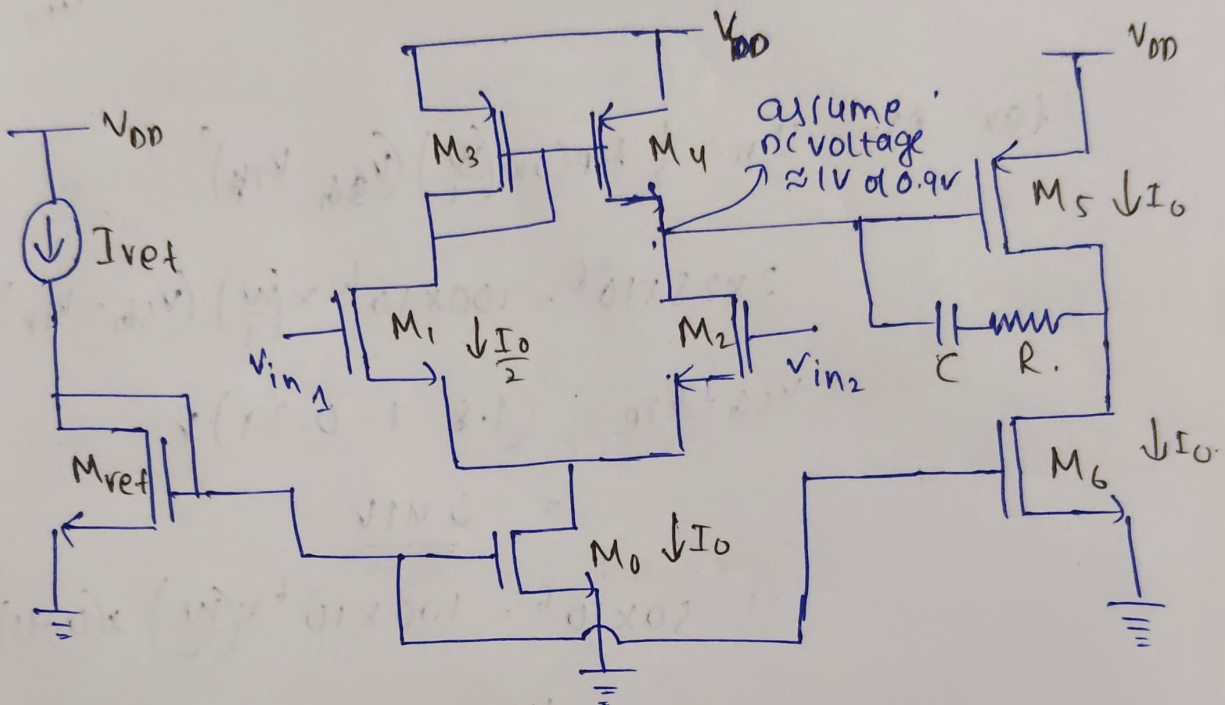


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## Two stage OTA design

→ let us choose  $I_{ref} = 5\mu A$  and tail current =  $50\mu A$



current to  $M_1 = 25\mu A$

assume  $100mV$  overdrive voltage for  $M_1, M_2$

$$\Rightarrow I_D = \frac{M_n C_{ox} W}{2L} \times (V_{GS} - V_{Th})^2$$

$$\text{given, } M_n C_{ox} = 230 \mu A / V^2$$

$$M_p C_{ox} = 100 \mu A / V^2$$

$$\Rightarrow 25\mu = \frac{1}{2} (230 \times 10^{-6}) \times \frac{W}{L} \times (0.1)^2$$

$$\Rightarrow \boxed{\frac{W}{L} \approx 21.74}$$

Choose  $L_1 = 1\mu$   $\Rightarrow$   $L_2 = 1\mu$   
 $W_1 = 21.74\mu$   $W_2 = 21.74\mu$

~~Choose~~  $M_{ref} = L = 1\mu$   $W = 10\mu$  (This is an assumption to ensure  $M_{ref}$  is in saturation)

$M_0$  will have  $\frac{W}{L}$  10 times  $I_{M_{ref}} \Rightarrow L_0 = 1\mu$   
 $W_0 = 100\mu$

for  $M_3$   $I_D = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L}\right) (V_{SG} - V_{TP})^2$

$$2 \times 25 \times 10^{-6} = 100 \times 10^{-6} \times \left(\frac{W}{L}\right) (V_{SG} - V_{TP})^2$$

$$V_{SG} - V_{TP} = (1.8 - 1 - 0.39)V$$

$$= \underline{\underline{0.41V}}$$

$$\Rightarrow 50 \times 10^{-6} = 100 \times 10^{-6} \times \left(\frac{W}{L}\right) \times (0.41)^2$$

$$\Rightarrow \underline{\underline{\frac{W}{L} = 2.974}}$$

Choose  $L_3 = 1\mu \Rightarrow W_3 = 2.974\mu$

$$\Rightarrow L_4 = 1\mu, W_4 = 2.974\mu$$

$M_6$  has same current as  $M_0$

$$\Rightarrow L_6 = 1\mu, W_6 = 100\mu$$

as we are using  $M_6$  current is  $50\mu A$

$\Rightarrow$  for  $M_5$  also same current flows

$$\Rightarrow \left(\frac{W}{L}\right)_5 = 2 \left(\frac{W}{L}\right)_3$$

$$\Rightarrow L_5 = 1\mu$$

$$W_5 = 5.984$$

theoretical  $g_{m1} = \frac{2I_1}{V_{GS} - V_{Th}} = \frac{2 \times 25 \times 10^{-6}}{10^{-1}}$

$$= 5 \times 10^{-4} \Omega^{-1}$$

simulated  $g_{m1} \approx 4.2 \times 10^{-4} \Omega^{-1}$

$$r_{o4} = \frac{1}{G_{ds4}} = \frac{1}{1 \times 10^{-6}} = 10^6 \Omega$$

$$r_{o2} = \frac{1}{G_{ds2}} = \frac{1}{2.99 \times 10^{-6}} = 3.3 \times 10^5 \Omega$$

} simulated

$$\Rightarrow r_{o2} \parallel r_{o4} = 2.48 \times 10^5$$

gain of stage 1 =  $g_{m1} (r_{o1} \parallel r_{o2})$

$$= 4.2 \times 10^{-4} \times 2.48 \times 10^5$$

$$= 104.16$$

$$\Rightarrow 20 \log(104.16) \approx 40 \text{ dB}$$

gain of stage-2:

$$g_{m5} = \frac{2I_5}{V_{SG} - V_{TP}} = \frac{2 \times 50 \times 10^{-6}}{1 - 0.418} \\ = \underline{1.71 \times 10^{-4} \Omega^{-1}}$$

simulated  $g_{m5} = 1.63 \times 10^{-4} \text{ A/V}$ .

$$r_{o5} = \frac{1}{g_{ds5}} = \frac{1}{2.39 \times 10^{-6}} = 4.18 \times 10^5$$
$$r_{o6} = \frac{1}{g_{ds6}} = \frac{1}{6.47 \times 10^{-6}} = 1.54 \times 10^5$$

} simulated.

$$\text{gain 2} = g_{m5} (r_{o5} \parallel r_{o6}) \\ \approx (1.63 \times 10^{-4}) (1.12 \times 10^5) \\ \approx \underline{18.256}$$

$$\text{net gain} = (\text{gain 1}) \times (\text{gain 2})$$

$$= 104.16 \times 18.256$$

$$= 1901.54$$

$\text{Overall gain} \approx 65.58 \text{ dB}$



we took  $C_2 = 5 \text{ pf}$

by Miller effect

$$\begin{aligned}(C_2)_{\text{input side}} &= (1+A) C_2 \\ &= (1+18.256) C_2 \\ &\approx \underline{\underline{96.28 \text{ pf}}}\end{aligned}$$

$$\begin{aligned}(C_2)_{\text{output side}} &= (1 \text{ pf}) + C_2 \\ &= \underline{\underline{6 \text{ pf}}}\end{aligned}$$

\* Theoretical

$$\text{Pole 1} \approx \frac{1}{R_{\text{eff}} (C_2)_{\text{input}}}$$

$$= \frac{1}{2.48 \times 10^5 \times 96.28 \text{ p}}$$

$$\approx \underline{\underline{41.8 \text{ kHz}}}$$

$$\text{Pole 2} \approx \frac{1}{(R_{\text{eff}}) C_{2 \text{ output}}}$$

$$= \frac{1}{1.12 \times 10^5 \times 6 \times 10^{-2}}$$

$$\approx 0.14 \times 10^7$$

$$\approx \underline{\underline{1400 \text{ kHz}}}$$

|                  | length ( $\mu\text{m}$ ) | width ( $\mu\text{m}$ ) |
|------------------|--------------------------|-------------------------|
| $M_{\text{ref}}$ | 14                       | 10                      |
| $M_0$            | 1                        | 100                     |
| $M_1$            | 1                        | 21.74                   |
| $M_2$            | 1                        | 21.74                   |
| $M_3$            | 1                        | 2.94                    |
| $M_4$            | 1                        | 2.94                    |
| $M_5$            | 1                        | 5.948                   |
| $M_6$            | 1                        | 100                     |

$$M_{\text{ref}}: V_{GS} - V_{TH} = 0.456 - 0.408 = 0.048$$

$$V_{DS} = 0.456$$

$$\Rightarrow V_{DS} > V_{GS} - V_{TH} \Rightarrow \text{Saturation}$$

$$\underline{M_0}: V_{GS} - V_{TH} = 0.456 - 0.409 = 0.047$$

$$V_{DS} = 0.316$$

$$V_{DS} > V_{GS} - V_{TH} \Rightarrow \text{saturation}$$

$M_1$ :

$$V_{GS} - V_{TH} \quad V_{DS}$$

$$\downarrow \quad \downarrow$$

$$0.584 - 0.5 \quad 0.5$$

$$\Rightarrow 0.5 > 0.084$$

$$\Rightarrow \text{Saturation}$$

$$M_2: V_{GS} - V_{Th} = 0.584 - 0.5 = 0.084$$

$$V_{DS} = 0.484$$

$$\Rightarrow V_{GS} - V_{Th} < V_{DS} \Rightarrow \text{saturation.}$$

$$M_3: V_{SG} - V_{Tp} < V_{SD}$$

$$0.897 - 0.417 < 0.974$$

$$\Rightarrow \text{saturation.}$$

$$M_4: V_{SG} - V_{Tp} < V_{SD}$$

$$0.897 - 0.417 < 0.974$$

$$\Rightarrow \text{saturation}$$

$$M_5: -V_{SG} - V_{Tp} = 1 - 0.418 = 0.582V.$$

$$V_{SD} = 0.904V$$

$$\Rightarrow \text{saturation.}$$

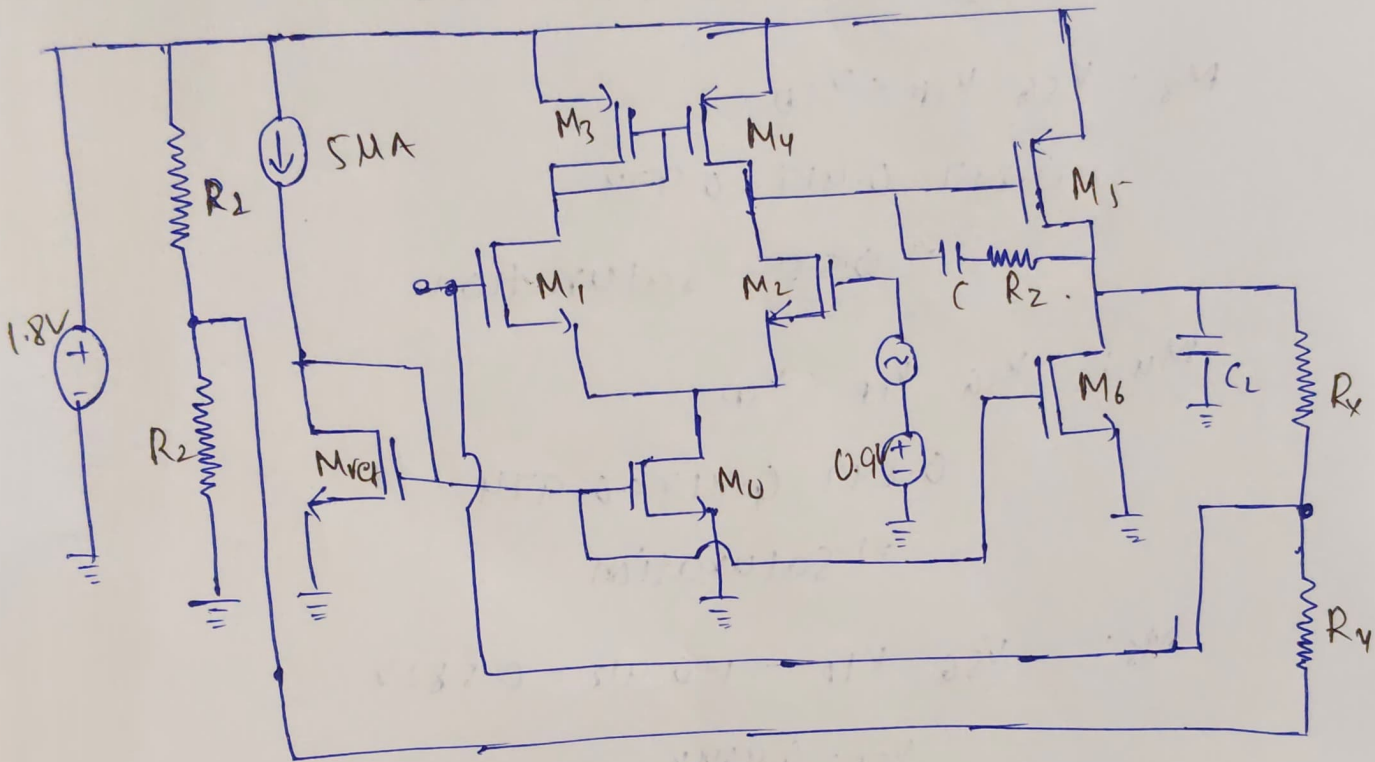
$$M_6: V_{GS} - V_{Th} = 0.456 - 0.405 = 0.05$$

$$V_{DS} = 0.899$$

$$\Rightarrow \text{saturation.}$$

$\Rightarrow$  All the Mosfets are in saturation.

so ~~the~~ finally the 2 stage ~~in~~OA placed in negative feed back.



used  $R_1, R_2$  as a voltage divider to get 0.9V, (choose  $R_x = R_y = 2000k$ )

Gain Bandwidth product  $\approx 900kHz$  (or)  $1MHz$ .

phase at 0dB =  $-103.5^\circ$

$\Rightarrow$  Phase Margin =  $76.5^\circ$

$\therefore$

$\Rightarrow$  Hence the system is stable.

So,  $M_4$

$C_1 = 13.55pF$

$C = 50.55pF$

$R_2 = 7.5kHz$

(These are obtained by fine tuning to get stable system)



## Power consumption:

$$\begin{aligned}\text{Total power consumption} &= V_{DD} \times I_{\text{total}} \\ &= V_{DD} \left( I_0 + I_0 + \frac{I_0}{10} \right) \\ &= 1.8 \text{ V} (50 + 50 + 5) \mu\text{A} \\ &= \underline{189 \mu\text{W}}\end{aligned}$$

## Slew rate:

The maximum rate of change of the output voltage with time is called slew rate.

$$\begin{aligned}\text{SR} &= \max \left( \frac{dV_{\text{out}}}{dt} \right) \\ &= \frac{I_{\text{tail}}}{C_c} = \frac{50 \mu\text{A}}{5 \text{ pF}} \\ &= 10 \text{ V}/\mu\text{s}\end{aligned}$$

## Input common mode range

$$V_{\text{inCM}} > V_{\text{GS1}}|_{I_0/2} + V_{\text{DSAT0}}$$

$$V_{\text{inCM}} < V_{DD} + V_{\text{Thn}} - V_{\text{TP}} - \sqrt{\frac{I_0}{K_{P3}}}$$

$$\Rightarrow \underline{0.8 < V_{\text{inCM}} < 1.6 \text{ V}}$$

for ~~my~~ pulse input of 0.2V.

~~0.2V~~ to

$$V_{\text{initial}} = 0$$

$$V_{\text{on}} = 0.2V$$

$$T_{\text{delay}} = 0.1\text{m}$$

$$T_{\text{rise}} = 0.1\text{n}$$

$$T_{\text{fall}} = 0.1\text{m}$$

small signal ac amplitude = 1

ac phase = 0

Gmin stepping succeeded in finding the operating point.

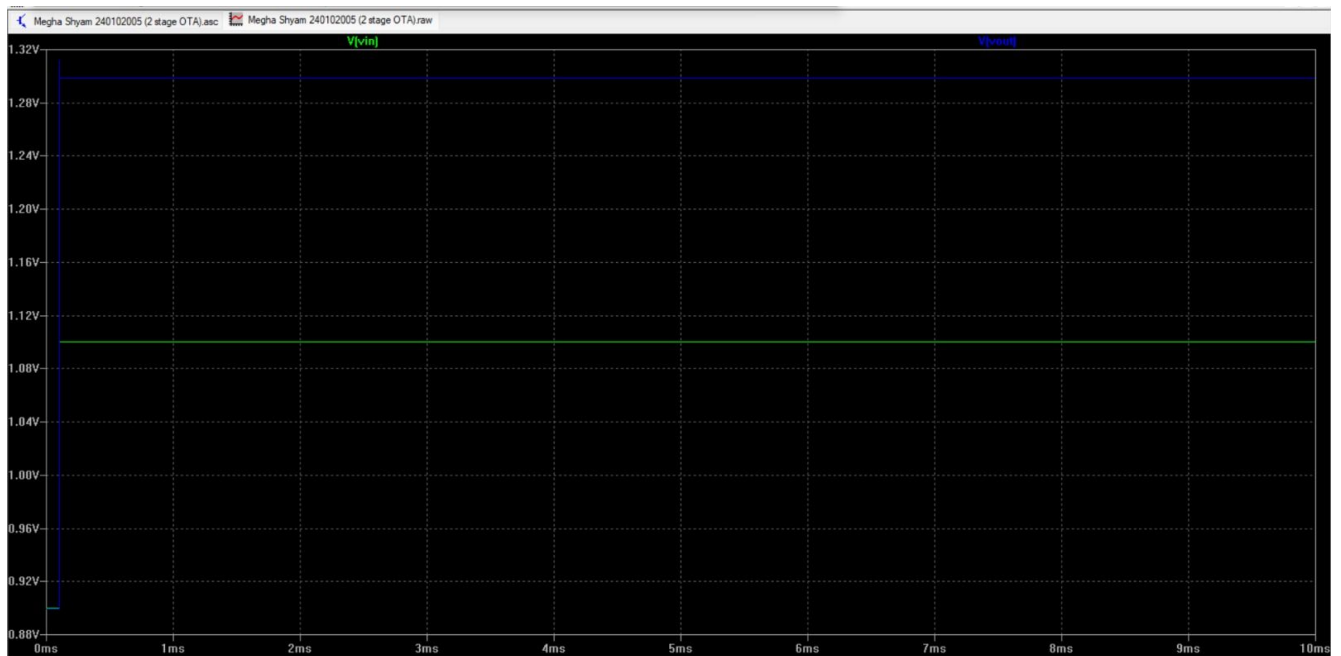
## Semiconductor Device Operating Points:

--- BSIM3 MOSFETS ---

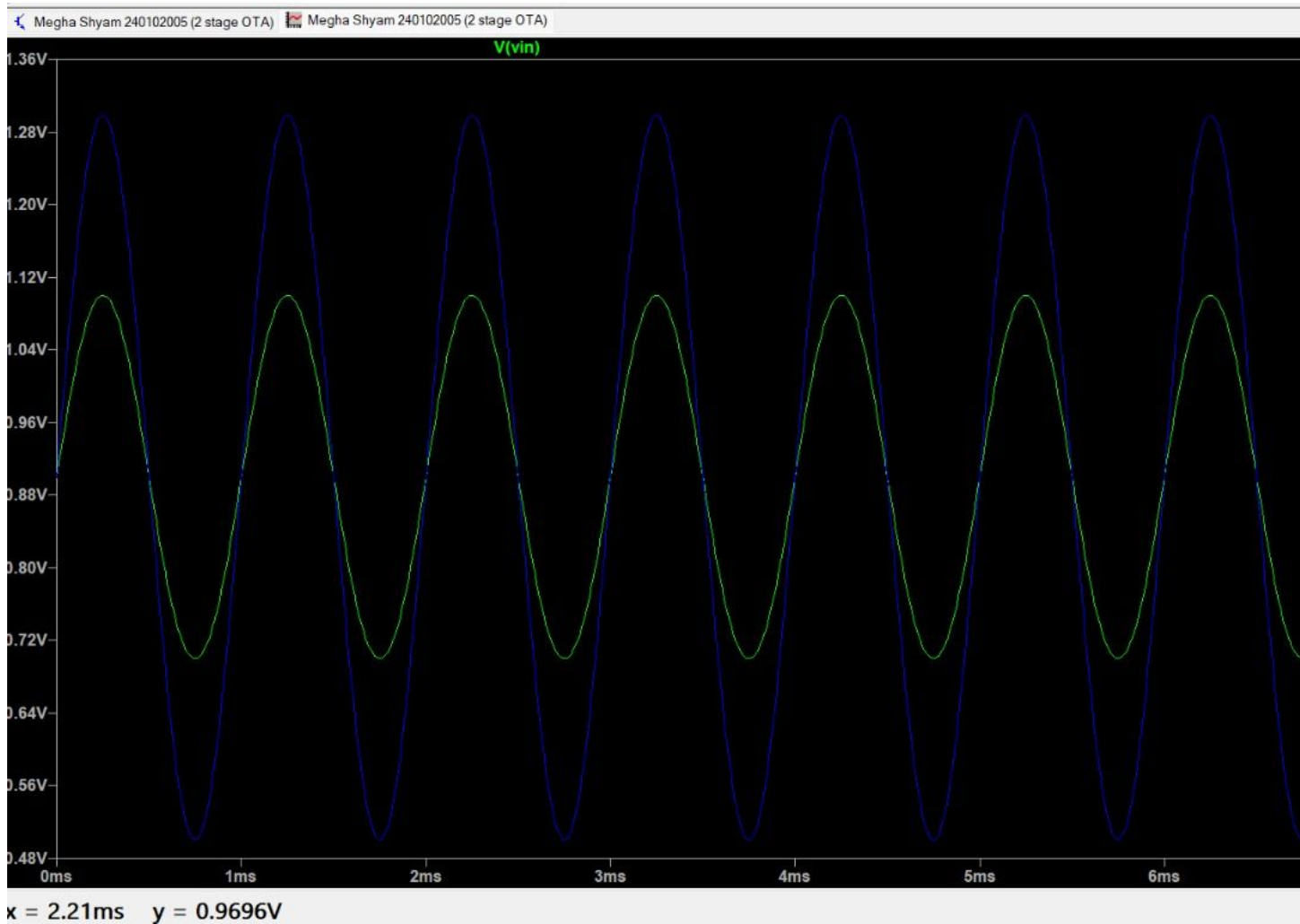
| Name:    | m5        | m4        | m3        | m2        | m1        |
|----------|-----------|-----------|-----------|-----------|-----------|
| Model:   | cmosp     | cmosp     | cmosp     | cmosn     | cmosn     |
| Id:      | -5.27e-05 | -2.44e-05 | -2.44e-05 | 2.44e-05  | 2.44e-05  |
| Vgs:     | -1.00e+00 | -9.74e-01 | -9.74e-01 | 5.84e-01  | 5.84e-01  |
| Vds:     | -9.01e-01 | -1.00e+00 | -9.74e-01 | 4.84e-01  | 5.10e-01  |
| Vbs:     | 0.00e+00  | 0.00e+00  | 0.00e+00  | -3.16e-01 | -3.16e-01 |
| Vth:     | -4.18e-01 | -4.17e-01 | -4.17e-01 | 5.00e-01  | 5.00e-01  |
| Vdsat:   | -4.38e-01 | -4.17e-01 | -4.17e-01 | 8.86e-02  | 8.85e-02  |
| Gm:      | 1.63e-04  | 7.94e-05  | 7.93e-05  | 4.20e-04  | 4.20e-04  |
| Gds:     | 2.39e-06  | 1.00e-06  | 1.02e-06  | 2.99e-06  | 2.93e-06  |
| Gmb:     | 5.34e-05  | 2.57e-05  | 2.56e-05  | 1.06e-04  | 1.06e-04  |
| Cbd:     | 0.00e+00  | 0.00e+00  | 0.00e+00  | 0.00e+00  | 0.00e+00  |
| Cbs:     | 0.00e+00  | 0.00e+00  | 0.00e+00  | 0.00e+00  | 0.00e+00  |
| Cgsov:   | 3.78e-15  | 1.89e-15  | 1.89e-15  | 1.79e-14  | 1.79e-14  |
| Cgdov:   | 3.78e-15  | 1.89e-15  | 1.89e-15  | 1.79e-14  | 1.79e-14  |
| Cgbov:   | 9.41e-19  | 9.41e-19  | 9.41e-19  | 9.66e-19  | 9.66e-19  |
| dQgdVgb: | 4.66e-14  | 2.33e-14  | 2.33e-14  | 1.73e-13  | 1.73e-13  |
| dQgdVdb: | -3.72e-15 | -1.86e-15 | -1.86e-15 | -1.75e-14 | -1.75e-14 |
| dQgdVsb: | -4.22e-14 | -2.11e-14 | -2.11e-14 | -1.44e-13 | -1.44e-13 |
| dQddVgb: | -1.95e-14 | -9.76e-15 | -9.76e-15 | -7.22e-14 | -7.22e-14 |

| Name:    | m6        | m0        | mref      |
|----------|-----------|-----------|-----------|
| Model:   | cmosn     | cmosn     | cmosn     |
| Id:      | 5.27e-05  | 4.87e-05  | 5.00e-06  |
| Vgs:     | 4.56e-01  | 4.56e-01  | 4.56e-01  |
| Vds:     | 8.99e-01  | 3.16e-01  | 4.56e-01  |
| Vbs:     | 0.00e+00  | 0.00e+00  | 0.00e+00  |
| Vth:     | 4.06e-01  | 4.09e-01  | 4.08e-01  |
| Vdsat:   | 6.57e-02  | 6.46e-02  | 6.48e-02  |
| Gm:      | 1.05e-03  | 9.86e-04  | 1.01e-04  |
| Gds:     | 6.47e-06  | 8.16e-06  | 7.06e-07  |
| Gmb:     | 3.05e-04  | 2.86e-04  | 2.92e-05  |
| Cbd:     | 0.00e+00  | 0.00e+00  | 0.00e+00  |
| Cbs:     | 0.00e+00  | 0.00e+00  | 0.00e+00  |
| Cgsov:   | 8.23e-14  | 8.23e-14  | 8.23e-15  |
| Cgdov:   | 8.23e-14  | 8.23e-14  | 8.23e-15  |
| Cgbov:   | 9.66e-19  | 9.66e-19  | 9.66e-19  |
| dQgdVgb: | 7.48e-13  | 7.44e-13  | 7.45e-14  |
| dQgdVdb: | -8.06e-14 | -8.10e-14 | -8.07e-15 |
| dQgdVsb: | -5.92e-13 | -5.85e-13 | -5.87e-14 |
| dQddVgb: | -3.00e-13 | -2.97e-13 | -2.98e-14 |
| dQddVdb: | 8.13e-14  | 8.17e-14  | 8.15e-15  |
| dQddVsb: | 2.88e-13  | 2.84e-13  | 2.85e-14  |
| dQbdVgb: | -1.49e-13 | -1.49e-13 | -1.50e-14 |
| dQbdVdb: | 2.16e-16  | -1.28e-16 | 1.08e-17  |

Transient response of the closed loop amplifier with a 0.2v step input.



Response of the closed loop amplifier with a 0.2v sinusoidal input.





## Frequency Response of the system

